

Algebra

[NEP]

Time: 2 Hours.

Full Marks: 40

The figures in the right-hand margin indicate marks.

Candidates are required to give their answer in their own words as far as practicable.

Symbols and notations have their usual meanings.

Answer all the questions.

Answer any five of the following questions. $2 \times 5 = 10$

- a) Find the product of all the values of $(1 + i\sqrt{3})^{\frac{3}{4}}$.
- b) If $d = \gcd(a, b)$, show that $\gcd(a^2, b^2) = d^2$.
- c) Solve in integers of the equation: $7x + 11y = 1$.
- d) Find the remainder when $1! + 2! + 3! + \dots + 2025!$ is divided by 15.
- e) Apply Descartes' rule of signs to examine the nature of the roots of the equations $x^4 + 2x^3 + 3x - 1 = 0$.
- f) Find the rank of the matrix $\begin{pmatrix} 0 & 0 & 2 & 2 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{pmatrix}$.
- g) If S is a real skew symmetric matrix of order n , prove that $(I_n + S)^{-1}(I_n - S)$ is orthogonal.
- h) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x, y, z) = (x + y, x - y, 0)$, $\forall (x, y, z) \in \mathbb{R}^3$. Show that T is a linear mapping.

[Turn Over]

. Answer any four of the following questions.

$5 \times 4 = 20$

- a) (i) Use the principle of mathematical induction to prove that $3^{4n+2} + 5^{2n+1}$ is divisible by 14 for all $n \in \mathbb{N}$.
(ii) Let $f, g: \mathbb{Z} \rightarrow \mathbb{Z}$ be two functions defined by $f(x) = [x], x \in \mathbb{R}$ and $g(x) = x + \frac{1}{2}, x \in \mathbb{Z}$. Find $f \circ g$ and $g \circ f$. 3+2
- b) (i) If $a, b, c, d > 0$ and $a + b + c + d = 1$, prove that $\frac{a}{1+b+c+d} + \frac{b}{1+a+c+d} + \frac{c}{1+a+b+d} + \frac{d}{1+a+b+c} \geq \frac{4}{7}$.
(ii) if n is a positive integer > 1 , prove that $n(n+1)^2 > 4(n!)^{3/n}$. 3+2
- c) (i) Find the $\arg z$, where $z = 1 + i \tan \frac{3\pi}{5}$
(ii) Let z be complex number and n be positive integer such that $(1+z)^n = p_0 + p_1 z + p_2 z^2 + \dots + p_n z^n$.
Prove that $p_0 - p_2 + p_4 - \dots = 2^{n/2} \cos \frac{n\pi}{4}$ and $p_1 - p_3 + p_5 - \dots = 2^{n/2} \sin \frac{n\pi}{4}$. 2+3
- d) Solve the equation $x^3 - 15x^2 - 33x + 847 = 0$ by Cardan's method.
- e) Find all eigen values of matrix $\begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{pmatrix}$. Hence find eigen vector corresponding to the greatest eigen value of this matrix. 3+2
- f) Let the matrix of a linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ relative to the ordered bases $\{(0,1,1), (1,0,1), (1,1,0)\}$ of $\mathbb{R}^3$ and $\{(1,0), (1,1)\}$ of $\mathbb{R}^2$ be $\begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 0 \end{pmatrix}$. Find T . Also find the matrix of T relative to the ordered bases $\{(1,1,0), (1,0,1), (0,1,1)\}$ of $\mathbb{R}^3$ and $\{(1,1), (0,1)\}$ of $\mathbb{R}^2$. 3+2

- g) (i) Define subspace of a vector space. Let S be the subset of $\mathbb{R}^3$ defined by $S = \{(x, y, z) \in \mathbb{R}^3 : 2x + y - z = 0\}$. Show that S is subspace of $\mathbb{R}^3$.

(ii) Apply $\gcd$ to find all units of $\mathbb{Z}_{18}$. 1+2+2

3. Answer *any one* of the following questions. **10 × 1 = 10**

- a) (i) Let z be a variable complex number such that

$$\left|z - \frac{10}{z}\right| = 3. \text{ Find the greatest and the least value of } |z|.$$

- (ii) Let $\rho = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} | a - b \text{ is a multiple of } 6\}$.

Show that ρ is an equivalence relation on $\mathbb{Z}$.

- (iii) Verify Cayley-Hamilton theorem for the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 3 & -1 \end{pmatrix}. \text{ Hence compute } A^{-1}. \quad 3+3+4$$

- (i) Apply elementary row operation to reduce the matrix

$$\begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{pmatrix} \text{ to a row echelon matrix.}$$

- (ii) Let $f: \mathbb{R} - \{1\} \rightarrow \mathbb{R} - \{1\}$ be defined by $f(x) = \frac{x+1}{x-1}$, $x \in \mathbb{R} - \{1\}$. Show that f is bijection. Determine f^{-1} .

- (iii) Solve the system of equations in integers:

$$x + 2y + z - 3w = 1$$

$$2x + 4y + 3z + w = 3$$

$$3x + 6y + 4z - 2w = 4. \quad 4+2+4$$
